

Alexandr Koryachko

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SKILLS

PROGRAMMING

Languages:

Python • PySpark • SQL • Matlab

Packages:

Pandas • Polars • Scikit-learn • PyTorch
numpy • NLTK • spaCy • mllib • Keras

Tools:

Git • Databricks • AWS • $\text{\LaTeX}$

DATA SCIENCE

Big Data • Statistics • Machine Learning

Data Visualization • Experiment Design

LINKS

LinkedIn:// [akoryachko](#)

Github:// [akoryachko](#)

GoogleScholar:// [bISiDH4AAAAJ](#)

EDUCATION

NORTH CAROLINA STATE UNIVERSITY

PHD IN ELECTRICAL ENGINEERING

Aug 2018 | Raleigh, NC

Minor in Mathematics

GPA: 4.0 / 4.0

NATIONAL RESEARCH UNIVERSITY OF ELECTRONIC TECHNOLOGY

MS IN COMPUTER SCIENCE

Jul 2009 | Moscow, Russia

Conc. in Software Engineering

GPA: 4.0 / 4.0

BS IN COMPUTER SCIENCE

Jun 2007 | Moscow, Russia

Conc. in Information Security

GPA: 3.92 / 4.0

COURSEWORK

Pattern Recognition

Detection and Estimation

Neural Networks

Mathematical Modeling

Uncertainty Quantification

Applied Bayesian Analysis

Random Processes

Graphs and Graphical Models

Computer Vision

Information Theory

Digital Signal Processing

Nonlinear Programming

Multivariable Control Systems

Numerical Analysis

Modeling of Biological Systems

Technical Communications

EXPERIENCE

XOMNIA | SENIOR DATA SCIENTIST

Feb 2025 - Present | Amsterdam, Netherlands

Consulting at Rabobank:

- Proposed and prototyped a model for predicting the next transaction date in periodic sequences.
- Standardized coding practices across team repositories, improving readability and reducing time spent on stylistic debates during PR reviews.
- Designed a monitoring framework that transformed a collection of notebook-based scripts into modular, object-oriented components that process only new data increments as they arrive.
- Organized knowledge-sharing sessions to equip fellow data scientists and engineers to use the developed tools effectively and independently.

Internal activity:

- Career mentor

XOMNIA | DATA SCIENTIST

Apr 2023 - Feb 2025 | Amsterdam, Netherlands

Consulting at Rabobank:

- Initiated and took a leading role in the periodicity cold start project. The project aimed to predict periodicity starting with the first transaction.
- Implemented a statistical test in PySpark on Databricks to ensure performance stability of the transaction periodicity detection model.

Internal activity:

- Authored a **blog post on data visualization** that ranked among the company's top 5 most-viewed articles for three consecutive months.
- Delivered a training session on PySpark best practices in Databricks.

VERICAST | STAFF DATA SCIENTIST I

Feb 2022 - Sep 2022 | Morrisville, NC, USA

Developed automated feature engineering functionality for big data applications within a scalable machine learning execution engine on AWS. The accompanying Python library was projected to save data scientists up to 90% of the time otherwise spent writing custom SQL queries for each new client dataset.

VERICAST | DATA SCIENTIST II

Mar 2021 - Feb 2022 | Morrisville, NC, USA

Prototyped tools for a privacy centered digital ad targeting solution that relies on webpage content as opposed to user tracking for placing clients' ads:

- Extended the optimization engine for digital advertisement auction bidding decisions with the support of cost per action campaigns.
- Built a classification model for identifying web pages with brand sensitive content. The model reduced ad serving on websites with unsafe content by 80% and improved clients' satisfaction with reporting.
- Built machine learning pipelines that analyze 300 million crawled web pages to select digital ad placements that are the most relevant to clients' desired content from 10 billion options daily.

VERICAST | DATA SCIENTIST I

Dec 2018 - Mar 2021 | Morrisville, NC, USA

Designed Machine Learning algorithms for Big Data processing

- Leveraged historical data on the outcomes of digital ad buying auctions to detect regions of flat win probability. This allowed saving about 2% on media costs by bidding at the lower side of such regions.
- Engineered tools for identifying store competitors from tens of millions of possible options by matching descriptions with spaCy Python library.
- Designed and tuned a random forest model for inventory quality prediction for digital ad auctions using Scikit-Learn.